WEEK-3

Java programming -Java Buzzwords, Data types, Variables, Operators, Control structures, Arrays, Console input and output

Sample Program:

1. Sample Java program to check whether the given number is prime palindrome or not. CO1.

2. Write a java program which reads marks of five subjects through command line and print the total and average.

Algorithm for the code:

Algorithm: Prime Palindrome

Step 1: Start.

Step 2: Read an integer n from the user.

Step 3: Check whether n is prime using isPrime(n).

To check prime:

1. **If n <= 1, return false.**
2. **Start a loop from i = 2 to n / 2.**
3. **Check n % i.**
4. **If n % i == 0, the number is not prime, so return false.**
5. **If no divisor is found, return true.**

Step 4: Check whether n is a palindrome using isPalindrome(n).

To check palindrome:

1. **Store the original number in original.**
2. **Set reversed = 0.**
3. **Extract the last digit using num % 10.**
4. **Add the digit to reversed.**
5. **Remove the last digit using num / 10.**
6. **Repeat until num becomes 0.**
7. **Compare original and reversed.**
8. **If they are equal, return true; otherwise return false.**

Step 5: Check both results using:

isPrime(n) && isPalindrome(n)

Step 6: If both are true, display:

n is a Prime Palindrome.

Otherwise, display:

n is NOT a Prime Palindrome.

Step 7: Close the Scanner.

Step 8: Stop.

3.UML diagram for the code:

```
┌──────────────────────────────────────┐
│                Week3                 │
├──────────────────────────────────────┤
│                                      │
│                                      │
├──────────────────────────────────────┤
│ + isPrime(num : int) : boolean       │
│                                      │
│ + isPalindrome(num : int) : boolean  │
│                                      │
│ + main(args : String[]) : void       │
└──────────────────────────────────────┘
```

4.Code:

```
import java.util.Scanner;

public class Week3 {

      public static boolean isPrime(int num) {
       if (num <= 1) return false;
       for (int i = 2; i <= num/2; i++) {
           if (num % i == 0) return false;
       }
       return true;
   }

   public static boolean isPalindrome(int num) {
       int original = num, reversed = 0;
       while (num > 0) {
           int digit = num % 10;
```

```
            reversed = reversed * 10 + digit;
            num /= 10;
        }
        return original == reversed;
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        if (isPrime(n) && isPalindrome(n)) {
            System.out.println(n + " is a Prime Palindrome.");
        } else {
            System.out.println(n + " is NOT a Prime Palindrome.");
        }
        sc.close();
    }
}
```

5.output:

```
Enter a number: 121
121 is NOT a Prime Palindrome.
PS C:\Users\Harsha Vardhan\Desktop\java> ^C
PS C:\Users\Harsha Vardhan\Desktop\java>
PS C:\Users\Harsha Vardhan\Desktop\java>  c:; cd 'c:\Users\Harsha Vardhan\Desktop\java'; & 'C:\Program Files\Java\jdk-25.0.4\bin\java.exe' '-X
X:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Harsha Vardhan\AppData\Roaming\Code\User\workspaceStorage\f8641d91bba67b5feb3c9719e0895
bbc\redhat.java\jdt_ws\java_61d938da\bin' 'Week3'
Enter a number: 131
131 is a Prime Palindrome.
PS C:\Users\Harsha Vardhan\Desktop\java>
```